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Less colorful = purer? The effect of packaging colorfulness on product purity perception

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ABSTRACT

This study examines how packaging colorfulness shapes consumer perceptions of product purity. Across five studies employing various stimuli, results demonstrate that consumers tend to perceive less colorful packaging as indicative of higher product purity compared to more colorful packaging. This perception bias arises from the inference that low colorfulness in packaging suggests fewer varieties of ingredients. Furthermore, the impact of packaging colorfulness on perceived product purity diminishes when detailed ingredient information is salient. These findings clarify how consumers infer product purity from the packaging colorfulness and provide strategic guidance for marketers in leveraging these perception biases for packaging design.

1. Introduction

Color is a fundamental visual design element in marketing. People form impressions within the first 90 s of encountering others or products, with approximately 62–90% of these judgments primarily shaped by color (Singh, 2006). Consequently, retailers frequently use packaging colors in retail environments to capture consumer attention and convey product features (Kauppinen-Räisänen, 2014; Martinez et al., 2021; Spence and Velasco, 2018; Ward et al., 2024). The famous jewelry brand, Tiffany & Co., prominently features its signature Tiffany Blue on packaging and promotional materials. This distinctive color embodies the brand's luxury, elegance, and individuality, helping its products stand out in a competitive jewelry market (TIFFANY & CO, 2024). Given thecritical role of packaging color in marketing, numerous researchers have explored the impact of color hue (Labrecque and Milne, 2012; Pelet and Papadopoulou, 2012; Tian and Huang, 2025), saturation (Hagtvedt and Adam Brasel, 2017; Huang et al., 2022; Liu et al., 2022; Mead and Richerson, 2018; Pichierri and Pino, 2023), and lightness (Hagtvedt, 2020; Jaud et al., 2025; Mai et al., 2016) on product judgments. These studies provide valuable insights into the effect of individual color dimensions. However, when considering color holistically, colorfulness emerges as a particularly influential factor. This study, therefore, seeks to explore how packaging colorfulness influences consumers' product judgments.

Colorfulness refers to the visual perception of an area displaying more or fewer colors, reflecting the vividness and diversity of colors visually presented within that space (Hunt, 1977; Song et al., 2022). Previous studies have examined how colorfulness influences consumer cognition and judgment in areas like advertising and branding. For example, Schindler (1986) found that the use of color in advertising not only captures attention but also enhances the realism and appeal of visual effects. Similarly, Song et al. (2022) observed that when consumers perceive a brand logo as more colorful, they often infer that the brand offers a diverse range of products. However, limited research has examined the impact of colorfulness on product attributes in the context of packaging. Therefore, this study aims to investigate how, why, and when packaging colorfulness influences consumers' perceived product purity.

Drawing from the spillover effect literature, which suggests that perceptions influenced by one stimulus may spill over onto perceptions of another (Balachander and Ghose, 2003; Janakiraman et al., 2009; Rutz and Bucklin, 2011), we hypothesize that the perception of fewer varieties, induced by low colorfulness in packaging, will spill over or transfer to consumers' judgments regarding the variety of ingredients inside, thus enhancing their perceptions of product purity.

Across five experiments, we established that packaging colorfulness influences consumers' purity judgments. This effect is driven by ingredient variety inference, where consumers infer that less colorful packaging contains fewer ingredient varieties, thereby heightening their perceived purity. Additionally, we propose that the salience of ingredient information moderates this effect. Specifically, the effect of packaging colorfulness on perceived product purity is attenuated when

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ingredient information is saliently provided.

This study offers several contributions to the literature in the relevant field. First, we identify colorfulness as a holistic visual element that encompasses an overall assessment of color and examine its impact on consumers' perceived product purity, thereby extending research on how packaging influences product perceptions (Gil-Pérez et al., 2020; Khan, 2020; Steiner and Florack, 2023). Additionally, we establish the mediating role of ingredient variety inference, thereby contributing to the literature on this topic (Kahn and Wansink, 2004; Townsend and Kahn, 2014). Finally, this study offers valuable practical insights for designers and marketers on how to leverage packaging color to shape purity perceptions, offering strategies to enhance consumer attitudes and judgments.

2. Theoretical background

2.1. Packaging color and product perception

Over the past decade, studies have demonstrated that aspects of packaging color, such as hue, saturation, and lightness, influence product perceptions (e.g., (Chung and Saini, 2022; Hagtvedt, 2020; Hagtvedt and Adam Brasel, 2017; Hallez et al., 2023; Pichierri and Pino, 2023; Sunaga et al., 2016). For example, Spence (2015) and Spence et al. (2010) demonstrated that altering the color hue of various products can significantly change the perceived taste. Hagtvedt and Adam Brasel (2017) found that enhancing color saturation captures consumers' attention, which subsequently triggers arousal and increases their size perceptions. Hagtvedt (2020) found that consumers perceive darker-colored products as heavier than lighter-colored ones, leading to judgments of greater durability but lower user-friendliness. These studies focus on individual specific dimensions of packaging color to examine differences in consumer product judgments. In contrast, our research explores the color effect from an overall color assessment perspective, focusing specifically on how packaging colorfulness influences consumer perceptions of the product inside.

2.2. Packaging colorfulness

Colorfulness is a visual perception outcome that determines the degree of different colors and can be influenced by the hue of specific colors and the number of colors used in the design (Song et al., 2022; Wang and Li, 2017). Therefore, presenting a variety of colors or enhancing the saturation of images can more effectively convey colorfulness (Scott and Vargas, 2007).

Research on colorfulness has investigated how this key color element influences consumer product judgments, particularly in the areas of product imagery, advertising, and brand logos. For instance, in the context of product imagery, Lee et al. (2014) explored the relationship between colorfulness and cognitive processing levels, finding that black-and-white (vs. colored) images encourage a higher level of construal, leading consumers to categorize objects according to characteristics associated with the level of construal. In the domain of advertising, Schindler (1986) found that the use of color in ads can better capture attention and make visual effects more realistic or appealing. Similarly, Meyers-Levy and Peracchio (1995) argued that when viewers devote fewer resources to processing ads, the use of color accentuates the appeal of surface cues, thereby facilitating the formation of consumer attitudes toward products. Furthermore, colorfulness plays a crucial role in brand logos. For instance, Yi et al. (2023) found that consumers exhibit a preference for colorful products and brand logos, particularly in response to health-related threats. The underlying mechanism is associated with death-related cognitions: health threats activate such thoughts, which in turn drive consumers to seek out colorfulness.

Building on previous research on colorfulness, this study examines an unexplored consequence of this key color element: its effect on consumers' product purity perceptions, thus contributing to research on factors that influence purity perceptions (Cho, 2019; Ton et al., 2024). We hypothesize that, compared to more colorful packaging, less colorful packaging will heighten consumers' perceived product purity.

2.3. Packaging colorfulness affects purity perception

Product purity is defined as the degree to which a product retains its essential characteristics, free from diluted or contaminated by non-essential ingredients (O'Hagan, 2020; Ton et al., 2024). For example, when a bottle of milk is perceived as pure, it implies that consumers believe the milk has not been adulterated by hormones, additives, or other non-essential ingredients. Similarly, a garment labeled as "pure cotton" typically suggests that the fabric consists entirely of cotton. In everyday consumer behavior, product purity is a key factor in how consumers evaluate products, shaping their attitudes and purchasing decisions (Li and Pandelaere, 2024; O'Hagan, 2020). We contribute to the literature on product purity by identifying packaging colorfulness as a novel visual driver.

Consumers often rely on external cues, such as packaging, advertising, and other marketing stimuli, to infer characteristics of a product or brand, even when these sensory cues appear to be unrelated to the product itself (Lowe and Haws, 2017; Park et al., 1986). In the current research, we propose that packaging colorfulness serves as an important visual cue that helps consumers judge product purity. On one hand, packaging with low colorfulness simplifies the visual design by reducing unnecessary embellishments and decorations, which promotes an impression of being free from impurities. This evaluation of the packaging extends to the product content, prompting consumers to associate packaging features with product features (Ackerman et al., 2010; Becker et al., 2011; Yu et al., 2022), thus perceiving the product as being untainted by excessive impurities. In contrast, packaging featured with various colors, due to its enhanced visual differences, may lead consumers to perceive more diverse elements in the packaging. These varied visual elements could evoke associations of unnecessary ingredients or impurities in the product, thereby negatively impacting their judgment of the product's purity.

On the other hand, packaging with low colorfulness typically features simple color patterns, which convey a sense of simplicity and unity visually, thereby evoking associations of minimal processing. This association is rooted in people's lay beliefs that more complex objects require more manual processing (Min et al., 2024). Specifically, packaging with simple colors (i.e., low colorfulness) reduces visual distractions and signals that the product maintains its most essential state, leading consumers to intuitively believe that the product contains fewer artificial ingredients and is less likely to be contaminated by unnecessary components (Wang et al., 2024). In contrast, more colorful packaging increases the differences between the elements in the packaging design, conveying a sense of sophistication. This makes consumers more likely to associate the product with human intervention or over-processing (Wang et al., 2024), thereby reducing the perception of the product's purity.

Collectively, less colorful packaging avoids excessive decoration and color, conveying an impression of no impurities and minimal processing. This impression of the packaging tends to transfer from the packaging to the products contained inside. As a result, consumers are likely to infer that packaging with less colorfulness contains purer products. More formally.

H1. The package with low (vs. high) colorfulness leads consumers to perceive the product inside as purer.

2.4. Spillover effects

The spillover effect refers to the transfer of affective or attribution evaluations, or both, from one entity to another related entity (Raufeisen et al., 2019). When the spillover effect occurs, existing perceptions and

information alter beliefs about attributes not directly linked to the perceived object (Ahluwalia et al., 2001; Janakiraman et al., 2009).

Previous studies have provided strong evidence for spillover effects across various domains. For example, research on brand attitudes has illustrated that a spillover effect exists between brands. Specifically, consumers' attitudes toward brand alliances can shape subsequent impressions of individual partner brands (Simonin and Ruth, 1998), and the perceptions of one brand can influence competing brands when they are sufficiently similar (Janakiraman et al., 2009). Research on emotions shows that when individuals enjoy background music, it evokes positive emotions and improves their product evaluations (Gorn, 1982). More relatedly, some scholars have extended the spillover effect to visual design. For instance, the sense of luxury evoked by visually artistic packaging may transfer to art-related products, resulting in a more luxurious evaluation of the product (Hagtvedt and Patrick, 2008). As another example, consumers transfer their overall preference evoked by the symmetry of a brand logo to their inference about the product design (i.e., aesthetics, functionality, and symbolism) (Bettels and Wiedmann, 2019). Spillover effects have been examined through various theoretical frameworks, including information integration theory, associative networks, and knowledge and attitude accessibility theories (Baumeister et al., 2015; Lei et al., 2008; Simonin and Ruth, 1998). All studies indicate that the spillover effect occurs in two stages: First, a cognitive connection is established between two entities; then, one entity's attributes are transferred or inferred onto the other entity (Raufeisen et al., 2019).

Establishing cognitive connections between two entities is a prerequisite for the spillover effect to occur. The stronger the connection between two entities in a consumer's memory and knowledge network, the more likely the spillover effect is to occur (Balachander and Ghose, 2003). Generally, to store and process information more effectively, individuals form cognitive connections between entities drawing on their past knowledge and experiences (Bar, 2004; Raufeisen et al., 2019). For example, when shopping, an individual may link a brand's products to recent advertisements, past experiences, or brand endorsers (Keller, 1993; Till and Busler, 2000).

Furthermore, attributes can be transferred between two entities that have formed mental connections. On one hand, due to time and cognitive constraints, the association between attributes encourages individuals to rely on available information and infer unavailable details to optimize cognitive efficiency (Ahluwalia et al., 2001). Particularly when certain attributes are unobservable, individuals are more likely to make inferences based on substitute information. For instance, the digital presence of service employees may affect consumers' evaluation of their service quality (Herhausen et al., 2020). On the other hand, when one entity is cognitively close to another and provides diagnostic information, it facilitates the transfer of existing observations from one entity to the other (Roehm and Tybout, 2006; Simonin and Ruth, 1998).

2.5. The spillover effect of packaging colorfulness on ingredient variety inference

To understand how packaging colorfulness influences product perceptions, it is essential to delineate the underlying factors driving the colorfulness-purity inference. We propose that the colorfulness-purity inference represents a specific form of the spillover effect, and it is important to identify the content of this colorfulness spillover. We theorize that the variety judgments induced by colorfulness may spill over. According to the Cambridge Dictionary, variety refers to the existence of multiple types or categories of things, serving as a characteristic that differentiates and diversifies items. Everyday experiences indicate that individuals frequently associate colorfulness with the sensory perception of variety. For instance, when observing a garden adorned with a multitude of colorful flowers, individuals naturally perceive the variety in nature. This association is further supported by empirical studies that confirmed a strong mental connection between colorfulness and the concept of variety (Post et al., 2017; Song et al., 2022). For instance, diversity in color or colorfulness in webpage design can create a greater sense of variety for users by accentuating the differences between elements (Post et al., 2017). Similarly, a greater variety of colors in food consumption indicates an increase in the diversity of nutrients being consumed (König, 2018; Penn Medicine, 2018). In summary, given the strong mental association between colorfulness and variety, we infer that colorfulness engenders a heightened perception of variety among individuals.

Building upon the clarification of the spillover content of colorfulness, we further examine how the two-step spillover effect unfolds. The first step involves establishing a cognitive connection between the packaging and the product it contains. In fact, packaging and the product it encloses are inherently connected. On one hand, the packaging is linked to the product it contains both temporally and spatially. Consumers' perceptions of both the packaging and the product arise almost simultaneously. Packaging and the product are perceived as inseparable by consumers until they are unwrapped and discarded. On the other hand, packaging consistently serves as an external representation of the product, conveying essential product information to consumers (Fajardo and Townsend, 2016). Packaging provides consumers with essential product details, including its category, weight, ingredients, and manufacturer. Extensive research has explored how packaging influences consumer perceptions of the product it contains (Felix et al., 2022; Madzharov and Block, 2010; Sevilla and Kahn, 2014; Ton et al., 2024; Veflen et al., 2023). Overall, the packaging is cognitively and deeply intertwined with the product it contains.

The second step of the spillover effect shows that perceptions can transfer between two entities that have already formed a mental connection. This implies that when visual cues on the packaging are linked to the product, the resulting perceptions and evaluations will spill over to judgments about the product (Huettl and Gierl, 2012). Therefore, we theorize that the sense of variety induced by packaging colorfulness will influence inferences about the ingredient variety of the product inside. Specifically, when consumers encounter packaging with lower colorfulness, they infer that the product is composed of fewer ingredient varieties. Collectively, drawing on psychological theories related to the spillover effect in the field of sensory marketing, we propose that packaging colorfulness fosters a sense of variety, which influences consumers' inferences about the ingredient variety within the packaging.

2.6. The ingredient variety inference affects purity perception

We hypothesize that consumers' inferences about ingredient variety influence their product purity perceptions. Specifically, when consumers perceive a product to contain fewer varieties of ingredients, they are more likely to view it as purer. This perception likely stems from an intuitive belief that products with fewer ingredients are less contaminated by non-essential components. Such a belief shapes consumers' overall product evaluations, leading them to perceive products with fewer ingredient varieties as embodying greater purity (Ton et al., 2024). For example, in product categories like food or cosmetics, consumers often perceive products with fewer ingredients as less likely to contain unnecessary additives, thereby deeming the products as purer and more natural (Schiano et al., 2021). More formally.

H2. The package with low (vs. high) colorfulness elicits a lower ingredient variety inference, which mediates the effect of packaging colorfulness on the purity perception of the product inside.

2.7. The moderating role of ingredient information salience

This study posits that consumers associate product packaging colorfulness with ingredient variety, leading them to infer that packaging with low (vs. high) colorfulness can reduce perceived ingredient

variety and enhance perceived product purity. However, we expect that the effect of colorfulness will be attenuated if ingredient information is available.

Ingredient information refers to the details provided on product packaging or promotional materials that outline the specific ingredients incorporated into the product. It offers consumers transparency about the product's composition, enabling them to understand the specific components of the product they are purchasing and using (Policastro et al., 2017). For example, an ingredient list is a common form of ingredient information, enumerating all raw materials, additives, and auxiliary substances included in the product, along with their respective quantities or proportions (Kraemer et al., 2023).

When ingredient information is saliently displayed on the packaging, consumers shift their attention from the packaging color to the ingredient details, enabling them to directly assess the actual variety of ingredients in the product (Chen et al., 2024). Consequently, consumers' perceptions of the product are recalibrated, and their judgment of ingredient variety no longer depends on packaging colorfulness but instead shifts to a more objective product evaluation (Newman et al., 2016). This shift prevents the sense of variety induced by colorfulness from spilling over onto consumers' inferences about ingredient variety. Specifically, ingredient information acts as a more reliable and compelling indicator of ingredient variety than packaging color. When ingredient information is clearly visible, consumers place greater emphasis on the actual ingredients, irrespective of the packaging's colorfulness level. In other words, consumers are less likely to directly associate packaging colorfulness with ingredient variety and will instead rely on objective ingredient information, minimizing perceptual biases driven by colorfulness. Collectively, when ingredient information is salient, consumers gain a clear understanding of the product's ingredients, reducing the likelihood of inferring ingredient variety and product purity solely based on packaging colorfulness. Therefore, the impact of packaging colorfulness on consumer purity perception may be diminished when ingredient information is salient. More formally.

H3. The effect of packaging colorfulness on consumers' perceived product purity is attenuated when ingredient information is salient.

3. Overview of studies

Fig. 1 illustrates our conceptual model. We tested our hypotheses across five experiments. We demonstrated that low (vs. high) packaging colorfulness enhanced consumers' perceived product purity, with ingredient variety inference identified as one potential mechanism. Studies 1A and 1B validated the basic effect of packaging colorfulness on product purity perception using two different manipulations of colorfulness (changing the variety or saturation of colors) (H1). Study 2 explored the underlying mechanism of this effect, testing the mediating role of ingredient variety inference (H2). Study 3 investigated the boundary condition of ingredient information salience, showing that the product purity perception induced by packaging colorfulness diminishes when ingredient information is salient (H3). Finally, Study 4 examined the downstream effect of packaging colorfulness such that less colorful packaging can increase product purity perception, which subsequently fosters purchase intentions.

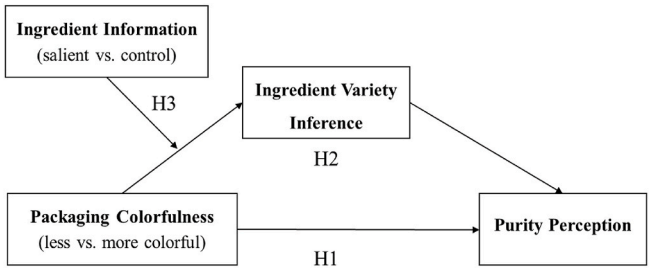


Fig. 1. Conceptual model.

We meticulously selected the stimuli for each study to ensure their relevance to the concept of product purity. The stimuli used in our experiments, including chocolate, soy milk, shampoo, as well as pure milk, are all products for which purity is a relevant and important attribute in consumers' perceptions. In the real market, many brands promote these food and personal care products as "pure" or "zero additives," with consumers tending to prefer purer products due to their association of purity with higher quality, minimal processing, and fewer artificial ingredients. These associations render the concept of purity particularly salient for the product categories we selected, ensuring the relevance and appropriateness of our stimuli.

The sample size for all experiments was pre-determined to be approximately 50 participants per condition (except for Study 4, which included about 100 participants per condition). Each study incorporated two attention check questions (e.g., " If you see a package of [product name] like this, please select 7 for the following question; " "Please do not select anything for the following question"). Only participants who passed these checks were counted in our predetermined sample size. Responses from participants who failed the attention checks were ineligible for payment and were not recorded by the data collection platform. Web Appendix A summarizes our results across all experiments.

4. Study 1A

Study 1A aimed to examine the main effect of packaging colorfulness on product purity perception (H1). We predicted that consumers would perceive higher product purity when encountering less (vs. more) colorful packaging.

4.1. Participants and design

Recruitment information was posted on Credamo (https://www.credamo.com), a widely recognized Chinese survey platform. One hundred and thirty-five participants participated in this study for a nominal payment. We employed a single factor (packaging colorfulness: low vs. high) between-subjects design. Participants were randomly assigned to either the low or high colorfulness condition. Considering the growing call among researchers for data cleaning (Chmielewski and Kucker, 2020; Ford, 2017), we referred to Yu et al.'s (2023) multiple-step cleaning protocol to further ensure data quality. This protocol was adhered to across all studies. Thirty-five participants who failed attention checks and one participant with an unknown IP address[1] were excluded from further analysis. As a result, we were left with 99 participants (M_{age} = 31.15, SD_{age} = 8.53; 70.7% female).

4.2. Procedure

Participants were informed that the study aimed to examine how consumers perceive food-related information. Under this pretense, participants imagined encountering a packet of black chocolates in a supermarket. They were shown an image of the chocolate. To manipulate the packaging colorfulness, we only varied in number of color types (Song et al., 2022). Specifically, in the low colorfulness condition, the packaging featured a single color, while in the high colorfulness condition, it included six distinct colors (i.e., blue, green, purple, orange, yellow, and pink) (see Appendix). Notably, the single color in the low colorfulness condition was randomly selected from the six colors used in the high colorfulness condition. Participants were then asked to rate their product purity perception using a three-item 7-point Likert scale adapted from Ton et al. (2024): "I think this product is pure," "I think this product is made of only essential ingredients," and "I think the

[1] We used IPHUB (https://iphub.info) to check IP addresses, and retained only participants whose IP addresses were marked as valid (i.e., "good IPs").

essential ingredients of this product are undiluted by other components." (1 = "strongly disagree," and 7 = "strongly agree"). Additionally, as a manipulation check for packaging colorfulness, participants completed a two-item scale adapted from Reinecke et al. (2013) and Song et al. (2022): "How colorful do you think this packaging is?" (1 = "not colorful at all," and 7 = "very colorful") and "To what extent this packaging has a complex color pattern?" (1 = "not complex at all," and 7 = "very complex"). It is worth noting that the concept of colorfulness was not mentioned until the manipulation check question, which was at the end of the questionnaire, so we could rule out the possibility that directing participants' attention to colorfulness led to a bias in their purity perception. We followed this rule in all studies. Finally, participants provided demographic information, including gender and age.

4.3. Results

Sample comparability. To verify the comparability of the two experimental conditions (n_{low} = 49; n_{high} = 50), we compared the gender and age of the participants in both conditions. Specifically, chi-square analysis revealed no significant difference in the gender distribution between the low colorfulness and high colorfulness conditions (χ^2 = .08, p = .775). An independent samples t-test showed no significant difference in age between participants in these two conditions (t(97) = .81, p = .418).

Manipulation check. The manipulation of packaging colorfulness was successful. Participants' responses to the two items were averaged (Cronbach's α = .71). An independent samples t-test showed that participants in the low colorfulness condition perceived the packaging as less colorful than those in the high colorfulness condition (M_{low} = 3.68, SD_{low} = 1.20 vs. M_{high} = 4.25, SD_{high} = 1.27; t(97) = −2.28, p = .025, d = −.46).

Product Purity Perception. We averaged participants' responses to the three items (Cronbach's α = .80). As hypothesized, an independent samples t-test showed that participants perceived the product inside as purer when presented with less colorful packaging compared to more colorful one (M_{low} = 5.18, SD_{low} = .99 vs. M_{high} = 4.31, SD_{high} = .89; t(97) = 4.65, p < .001, d = .93), which supported H1.

4.4. Discussion

Preliminary evidence from Study 1A supports our hypothesis that packaging colorfulness influences consumers' product purity perception. Consistent with our expectations, participants perceived the product as purer when presented with low colorfulness. However, we noted that in Study 1A, colorfulness was manipulated by altering the number of color types, resulting in an inconsistency in the number of products displayed between the high and low colorfulness conditions. Specifically, in the high colorfulness condition, one pack of dark chocolate with six colors was presented, whereas in the low colorfulness condition, six packs with single-color packaging were presented. This issue will be addressed in subsequent studies.

5. Study 1B

Study 1B sought to test our basic hypothesis (H1) by using a different manipulation of colorfulness and a distinct product stimulus.

5.1. Participants and design

One hundred and nineteen participants from Credamo.com participated in this study for payment. A between-subjects design with a single factor (packaging colorfulness: low vs. high) was used. Participants were randomly assigned to either the low or high colorfulness condition. Nineteen participants who failed attention checks were excluded from further analysis. As a result, we were left with 100 participants (M_{age} = 31.44, SD_{age} = 7.81; 76.0% female).

5.2. Procedure

Participants were instructed to imagine seeing a bag of soy milk in a supermarket. They were then shown an image of the soy milk. Given that colorfulness can be not solely conveyed by the presence of multiple colors, but also by highly saturated colors in images (Lee et al., 2014; Scott and Vargas, 2007; Song et al., 2022), we manipulated colorfulness by altering the color saturation. Specifically, in the low colorfulness condition, the soya milk package exhibited a low saturation pattern, while in the high colorfulness condition, it displayed a highly saturated pattern (see Appendix). To ensure that the two conditions differed only in saturation and were identical in all other aspects, we used Photoshop's saturation adjustment tool to decrease the saturation parameter of the packaging pattern in the high colorfulness condition by 90%, creating a low colorfulness version. This manipulation method was used in the following studies. After viewing the soy milk package, participants reported their product purity perceptions and completed manipulation check questions regarding packaging colorfulness, consistent with Study 1A. Finally, participants provided demographic information, including gender and age.

5.3. Results

Sample comparability. To verify the comparability of the two experimental conditions (n_{low} = 50; n_{high} = 50), we compared the gender and age of the participants in both conditions. Specifically, chi-square analysis revealed no significant difference in the gender distribution between the low colorfulness and high colorfulness conditions (χ^2 = 1.97, p = .160). An independent samples t-test showed no significant difference in age between participants in these two conditions (t(98) = −.23, p = .819).

Manipulation check. The manipulation of packaging colorfulness was successful. We averaged participants' responses to the two items (Cronbach's α = .74). Participants in the low colorfulness condition perceived the packaging as less colorful than those in the high colorfulness condition (M_{low} = 4.23, SD_{low} = 1.40 vs. M_{high} = 5.34, SD_{high} = 1.07; t(98) = −4.45, p < .001, d = −.89).

Product Purity Perception. We averaged participants' responses to the three items (Cronbach's α = .85). Participants perceived the product as purer when they saw packaging with low colorfulness compared to high colorfulness (M_{low} = 4.79, SD_{low} = 1.25 vs. M_{high} = 3.98, SD_{high} = 1.07; t(98) = 3.49, p < .001, d = .70), which reaffirmed H1.

5.4. Discussion

The results of Study 1B again supported H1, showing that the colorfulness of product packaging influenced consumers' product purity perceptions. Additionally, we extended our experimental stimuli from solid food (dark chocolate) to liquid food (soy milk). More importantly, we used a different manipulation method from Study 1A, which varied color types, by adjusting color saturation to manipulate colorfulness. These results further validated the robustness and generalizability of our findings. However, the underlying mechanism remains unclear, thus our next study (Study 2) will explore how ingredient variety inference mediates this effect.

6. Study 2

Study 2 was designed with two main objectives in mind. First, while previous studies used edible consumable products (dark chocolate and soy milk) as stimuli, this study tested whether the colorfulness effect could be generalized to a nonedible consumable product (shampoo). Furthermore, this study aimed to explore the underlying mechanism by examining the mediating role of ingredient variety inference (H2). We predicted that packaging with low colorfulness would lead to a higher perception of product purity, as consumers would infer that the product

contained fewer ingredient varieties.

6.1. Participants and design

One hundred and twenty-three participants from Credamo.com participated in a between-subjects paid study with one factor: packaging colorfulness (low vs. high). Participants were randomly assigned to either the low or high colorfulness condition. Twenty-three participants who failed attention checks were excluded from further analysis. As a result, we were left with 100 participants (M_{age} = 31.00, SD_{age} = 7.91; 76.0% female).

6.2. Procedure

Participants were asked to imagine encountering a bottle of shampoo in a supermarket. They were shown an image of the shampoo. Packaging colorfulness was manipulated by using either a low or high color saturation pattern on the shampoo (see Appendix). As in previous studies, participants reported their product purity perceptions. Next, we measured ingredient variety inference using a three-item scale: "How many types of ingredients do you think this product contains?"; "How many types of ingredients do you think this product is mixed with?"; "How many types of ingredients do you think this product is made of?" (1 = "very few", 7 = "very many"; (Ton et al., 2024). Participants also completed the same manipulation check questions as in previous studies. Finally, participants provided demographic information, including gender and age.

6.3. Results

Sample comparability. To verify the comparability of the two experimental conditions (n_{low} = 50; n_{high} = 50). we compared the gender and age of the participants in both conditions. Specifically, chi-square analysis revealed that there was no difference in the gender distribution between the low colorfulness and high colorfulness conditions (χ^2 = .22, p = .640). An independent samples t-test showed that there was no difference in age between participants in these two conditions ($t(98)$ = 1.32, p = .190).

Manipulation check. The manipulation of packaging colorfulness was successful. We averaged participants' responses to the two items (Cronbach's α = .72). Results showed that participants in the low colorfulness condition perceived the packaging as less colorful than those in the high colorfulness condition (M_{low} = 3.60, SD_{low} = 1.25 vs. M_{high} = 5.84, SD_{high} = .77; $t(98)$ = −10.81, p < .001, d = −2.16).

Product Purity Perception. We averaged participants' responses to the three items (Cronbach's α = .85). Participants perceived the product as purer when the packaging was low in colorfulness compared to high colorfulness (M_{low} = 5.06, SD_{low} = 1.07 vs. M_{high} = 3.69, SD_{high} = 1.20; $t(98)$ = 6.03, p < .001, d = 1.21).

Ingredient Variety Inference. We averaged participants' responses to the three items (Cronbach's α = .95). Participants who saw less colorful packaging perceived the product to contain fewer ingredient varieties than those who saw more colorful one (M_{low} = 4.17, SD_{low} = 1.51 vs. M_{high} = 5.51, SD_{high} = 1.08, $t(98)$ = −5.12, p < .001, d = −1.02). These results suggested that packaging colorfulness influenced consumers' inferences about the variety of ingredients in the product.

Mediation Analysis. The results from PROCESS model 4 (5000 samples; Hayes, 2012) confirmed the proposed mediation model in which packaging colorfulness (low = 1, high = 0) was the independent variable, ingredient variety inference was the mediator, and purity perception was the dependent variable. As Fig. 2 indicated, packaging colorfulness significantly influences ingredient variety inference (b = −1.34, SE = .26; t = −5.12, p < .001). When both the independent variable (packaging colorfulness) and the mediator (ingredient variety inference) were used to predict purity perception, we observed a significant effect of ingredient variety inference (b = −.48, SE = .07; t = −6.57, p < .001) and the effect of packaging colorfulness also remained significant (b = .73, SE = .21; t = 3.40, p = .001). Moreover, results revealed that there is a mediated relationship between the packaging colorfulness and product purity perception through ingredient variety inference (b = .65, SE = .17; 95% CI: [.3483, 1.0039]), which implied that the effect is partially mediated by ingredient variety inference (Zhao et al., 2010). Study 2 results affirmed H2, highlighting the role of ingredient variety inference in mediating the effect of packaging colorfulness on perceived purity.

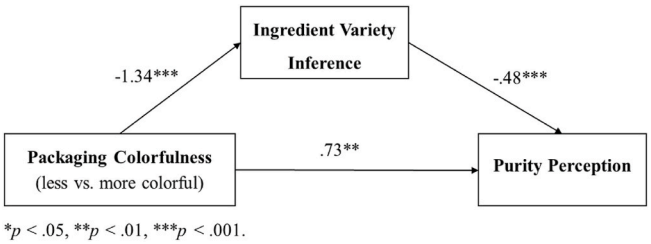


Fig. 2. Mediation model (Study 2).

6.4. Discussion

Study 2 replicated previous findings, extending the effect of packaging colorfulness on perceived product purity from the edible to the nonedible consumable product category. More importantly, this study explored the underlying process of this colorfulness effect by examining the mediating role of ingredient variety inference. Specifically, participants inferred that packaging with low colorfulness contained fewer ingredient varieties, leading to higher perceived product purity. Overall, these findings deepen our understanding of how consumers respond to varying colorfulness levels, emphasizing the mediating role of ingredient variety inference. However, does the colorfulness effect apply in all situations? This question calls for further exploration of its boundary conditions. Therefore, in Study 3, we will examine the moderating role of ingredient information salience to gain a deeper understanding of the main effect and its underlying mechanisms.

7. Study 3

The purpose of Study 3 was twofold. First, to enhance ecological validity, we selected a real product available on Amazon as the stimulus. Second, this study examined the moderating effect of ingredient information salience on the relationship between packaging colorfulness and perceived purity (H3), further reinforcing ingredient variety inference as a potential mechanism. We predicted that when ingredient information was salient, the colorfulness effect on purity perception would diminish.

7.1. Participants and design

Two hundred and twenty-one participants from Credamo.com participated in Study 3 in exchange for payment. Participants were randomly assigned to one of four conditions in a 2 (packaging colorfulness: low vs. high) × 2 (ingredient information: salient vs. control) between-subjects design. Twenty-one participants who failed attention checks and one participant whose IP was unknown were excluded from further analysis. As a result, we were left with 199 participants (M_{age} = 32.75, SD_{age} = 9.17; 67.3% female).

7.2. Procedure

Participants were asked to imagine encountering a chocolate in a supermarket. The chocolate image was presented to the participants. A real product (Tony's chocolonely) image was selected from Amazon, a leading online shopping platform. The package featured a highly

colorful design. As in previous studies, Photoshop was used to decrease the saturation parameter of the packaging pattern in the high colorfulness condition, thereby creating a low colorfulness version (see Appendix). Based on prior literature (Sah et al., 2021; Ton et al., 2024), we manipulated ingredient information salience. Specifically, in the salient condition, participants viewed the chocolate's ingredient details (i.e., a detailed list of all the raw materials, additives, and auxiliary substances included in the product) to participants, while no such information was provided in the control condition (see Appendix). Notably, to enhance ecological validity, we used the actual ingredient list of this chocolate (https://shop.frankfurt-airport.com/en-US/product/tony-s-hun-he-qiao-ke-li-dai-zhuang-320g_79fff3da623eddf0). After viewing the information, participants rated perceived product purity, ingredient variety inference, and completed a manipulation check regarding packaging colorfulness, consistent with prior studies. Additionally, we checked whether the manipulation of ingredient information salience was successful by asking, "In the above material, did you see the ingredient list of this bag of chocolate?" (1 = "yes," and 0 = "no"). This question was placed at the end of the questionnaire, to ensure it did not influence participants' responses to the main experimental tasks. Finally, participants were asked to provide demographic information, including gender and age.

7.3. Results

Sample comparability. We verified the comparability of the two experimental conditions with different levels of colorfulness when ingredient information is salient ($n_{low} = 50$; $n_{high} = 50$) and when it is not salient ($n_{low} = 50$; $n_{high} = 49$). Specifically, there was no difference in gender distribution between the low colorfulness and high colorfulness conditions, both when the ingredient information is salient ($\chi^2 = .74$, $p = .391$) and not salient ($\chi^2 = .08$, $p = .776$). Participants in these two colorfulness conditions showed no difference in age, both when the ingredient information is salient ($F(1, 195) = 1.16$, $p = .283$) and not salient ($F(1, 195) = .03$, $p = .868$).

Manipulation check. The manipulation check for packaging colorfulness was successful. Specifically, in both the salient and control conditions, participants rated the packaging as less colorful in the low colorfulness condition compared to the high colorfulness condition (Salient condition: $M_{low} = 4.63$, $SD_{low} = 1.25$ vs. $M_{high} = 5.93$, $SD_{high} = .72$; $F(1, 195) = 39.60$, $p < .001$, $\eta_p^2 = .17$; Control condition: $M_{low} = 4.27$, $SD_{low} = 1.33$ vs. $M_{high} = 6.27$, $SD_{high} = .64$; $F(1, 195) = 92.34$, $p < .001$, $\eta_p^2 = .32$).

Furthermore, the manipulation of ingredient information salience was successful. The chi-square analysis revealed that, in both the low and high colorfulness conditions, more participants noticed the ingredient information when it was salient compared to when it was not salient (Low colorfulness: 88.0% vs. 10.0%: $\chi^2 = 60.86$, $p < .001$; Odds ratio = 66.00; High colorfulness: 96.0% vs. 2.0%: $\chi^2 = 87.40$, $p < .001$; Odds ratio = 1152.00).

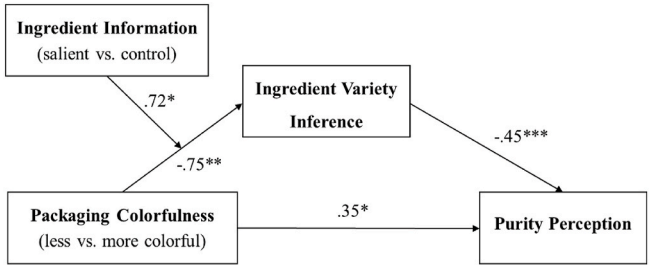
Product Purity Perception. We averaged participants' responses to the three items (Cronbach's α = .83). A two-way ANOVA revealed a significant main effect of packaging colorfulness ($M_{low} = 4.45$, $SD_{low} = 1.35$ vs. $M_{high} = 3.92$, $SD_{high} = 1.38$; $F(1, 195) = 7.93$, $p = .005$, $\eta_p^2 = .04$), a marginally significant main effect of ingredient information salience ($M_{salient} = 4.02$, $SD_{salient} = 1.50$ vs. $M_{control} = 4.35$, $SD_{control} = 1.25$; $F(1, 195) = 2.86$, $p = .093$, $\eta_p^2 = .01$), and a significant interaction effect ($F(1, 195) = 11.20$, $p < .001$, $\eta_p^2 = .05$). Planned contrast showed that when ingredient information is not salient, participants' purity perception was significantly higher when product packaging featured a low colorfulness pattern compared to a high colorfulness pattern ($M_{low} = 4.92$, $SD_{low} = .97$ vs. $M_{high} = 3.76$, $SD_{high} = 1.25$; $F(1, 195) = 18.89$, $p < .001$, $\eta_p^2 = .09$), which is consistent with results from prior studies. However, when the ingredient information was salient, there was no difference in consumers' product purity perception between the high colorfulness condition and the low colorfulness condition ($M_{low} = 3.97$, $SD_{low} = 1.52$ vs. $M_{high} = 4.07$, $SD_{high} = 1.49$; $F(1, 195) = .14$, $p = .706$, $\eta_p^2 = .00$).

Ingredient Variety Inference. We averaged participants' responses to the three items (Cronbach's α = .94). A two-way ANOVA revealed a significant main effect of packaging colorfulness ($M_{low} = 5.10$, $SD_{low} = 1.46$ vs. $M_{high} = 5.50$, $SD_{high} = 1.25$; $F(1, 195) = 5.67$, $p = .018$, $\eta_p^2 = .03$) and ingredient information salience ($M_{salient} = 5.98$, $SD_{salient} = .93$ vs. $M_{control} = 4.62$, $SD_{control} = 1.41$; $F(1, 195) = 67.08$, $p < .001$, $\eta_p^2 = .26$), and also a significant interaction effect ($F(1, 195) = 4.75$, $p = .031$, $\eta_p^2 = .02$). Planned contrasts revealed that, under the control condition, participants' ingredient variety inference was significantly lower for the low colorfulness packaging compared to the high colorfulness packaging ($M_{low} = 4.25$, $SD_{low} = 1.39$ vs. $M_{high} = 5.00$, $SD_{high} = 1.33$; $F(1, 195) = 10.34$, $p = .002$, $\eta_p^2 = .05$). However, under the salient condition, no difference was observed in ingredient variety inference between the high and low colorfulness conditions ($M_{low} = 5.96$, $SD_{low} = .92$ vs. $M_{high} = 5.99$, $SD_{high} = .94$; $F(1, 195) = .02$, $p = .886$, $\eta_p^2 = .00$).

Moderated Mediation. We conducted a moderated mediation analysis in which packaging colorfulness (low = 1, high = 0) was the independent variable, ingredient variety inference was the mediator, purity perception was the dependent variable, and ingredient information salience (salient = 1, control = 0) was the moderator (PROCESS Model 7; Hayes, 2012). As Fig. 3 showed, packaging colorfulness significantly influenced ingredient variety inference (b = −.75, SE = .23; $t = -3.22$, $p = .002$). The interaction effect of packaging colorfulness and ingredient information salience on ingredient variety inference was significant (b = .72, SE = .33; $t = 2.18$, $p = .031$). Furthermore, ingredient variety inference negatively influenced product purity perception (b = −.45, SE = .06; $t = -6.91$, $p < .001$). More importantly, the results revealed a significant index of moderated mediation (b = −.32, SE = .15; 95% CI = [-.6406, −.0339]), and a significant conditional indirect effect of packaging colorfulness on purity perception through ingredient variety inference when the ingredient information is not salient (b = .34, SE = .13; 95% CI = [.0942, .6013]), but not when the ingredient information is salient (b = .01, SE = .08; 95% CI = [-.1527, .1736]).

7.4. Discussion

The results of Study 3 supported H3, suggesting that ingredient information salience moderates the effect of packaging colorfulness on consumers' purity perception. Specifically, when ingredient information was salient, no difference in consumers' perceived product purity was observed, regardless of packaging colorfulness. However, this study only considered the situation where the ingredient information was provided as an intuitive and complete list of ingredients. In real-world markets, consumers are also exposed to implied ingredient information, such as nutrition claims that mention a specific nutrient (e.g., "rich in vitamin C," "low in fat," and "no added sugar") without providing detailed ingredient lists (Belei et al., 2012). This raises an important question: when ingredient information is implied rather than explicitly detailed, does it still moderate the effect of packaging colorfulness on product



*p < .05, **p < .01, ***p < .001.

Fig. 3. Moderated mediation model (Study 3).

purity perception? To address this question, we conducted Supplementary Study 1 (see Web Appendix B), which revealed that when the implied ingredient information was present, the effect of packaging colorfulness on perceived purity diminished.

Moreover, most previous studies manipulated packaging colorfulness by adjusting saturation. Given that packaging with extremely desaturated colors is highly atypical in a retail context and unlikely to be encountered on supermarket shelves, the next study (Study 4) manipulated colorfulness by presenting either a single-colored or a multi-colored package.

8. Study 4

Study 4 was conducted to reaffirm H1 and H2, by manipulating single-colored versus multi-colored packaging designs. More importantly, this study aimed to test the downstream consequence of the colorfulness-purity effect. Existing research has shown that consumers often value purity, resulting in more positive attitudes and behavioral outcomes toward products with high purity (Li and Pandelaere, 2024; Ton et al., 2024). In this study, we predicted that less colorful packaging would result in greater purity perception, thereby enhancing purchase intentions.

8.1. Participants and design

Two hundred and sixty-eight participants from Credamo.com participated in Study 4 in exchange for payment. Participants were randomly assigned to one of two conditions in a single factor (packaging colorfulness: low vs. high) between-subjects design. Sixty-nine participants who failed attention checks were excluded from further analysis. As a result, we were left with 199 participants (M_{age} = 31.64, SD_{age} = 7.71; 72.9% female)

8.2. Procedure

Participants were asked to imagine seeing a box of milk in a supermarket. An image of the milk was presented to them. We manipulated colorfulness by varying the number of color types in the packaging design. Specifically, in the low colorfulness condition, the packaging featured a single color (turquoise), whereas in the high colorfulness condition, the packaging featured multiple colors, including turquoise, orange, blue, green, brown, and a total of 12 colors (see Appendix). After viewing the box of milk, participants were asked to report their purchase intentions ("How likely are you to purchase this box of milk if you need it?" 1 = "not at all likely" and 7 = "very likely"; Rodas and John, 2020). As in previous studies, participants reported their product purity perceptions and ingredient variety inference. Participants also completed the same manipulation check questions used in previous studies. Finally, participants provided demographic information, including gender and age.

8.3. Results

Sample comparability. To verify the comparability of the two experimental conditions (n_{low} = 100; n_{high} = 99), we compared the gender and age of participants in both groups. Specifically, there was no difference in gender distribution between the low colorfulness and high colorfulness conditions (χ^2 = .00, p = .965). Participants in these two conditions showed no difference in age (t(197) = .89, p = .377).

Manipulation check. The manipulation of packaging colorfulness was successful. Results showed that participants in the low colorfulness condition perceived the packaging as less colorful than those in the high colorfulness condition (M_{low} = 3.18, SD_{low} = 1.12 vs. M_{high} = 4.36, SD_{high} = 1.19; t(197) = −7.22, p < .001, d = −1.02).

Purchase Intention. Results showed that, compared to milk inside more colorful packaging, participants exhibited a stronger purchase intention toward milk inside less colorful packaging (M_{low} = 5.55, SD_{low} = .91 vs. M_{high} = 4.89, SD_{high} = 1.21, t(197) = 4.35, p < .001, d = .62).

Product Purity Perception. We averaged participants' responses to the three items (Cronbach's α = .81). Participants perceived the product as purer when the packaging was low in colorfulness compared to high colorfulness (M_{low} = 5.41, SD_{low} = .83 vs. M_{high} = 4.44, SD_{high} = 1.12; t(197) = 6.90, p < .001, d = .98).

Ingredient Variety Inference. We averaged participants' responses to the three items (Cronbach's α = .93). Participants who saw less colorful packaging perceived the product to contain fewer ingredient varieties than those who saw more colorful one (M_{low} = 3.13, SD_{low} = 1.26 vs. M_{high} = 4.32, SD_{high} = 1.51, t(198) = −6.05, p < .001, d = −.86).

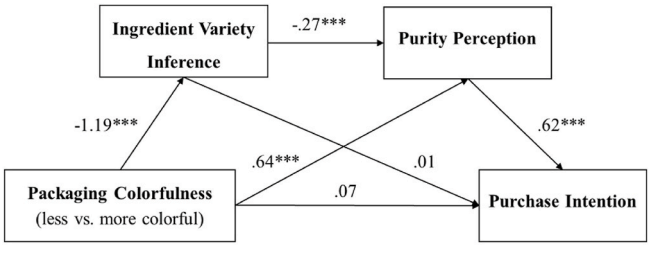
Serial Mediation Analysis. To test the full causal chain, we conducted a serial mediation analysis, with packaging colorfulness (low = 1, high = 0) as the independent variable, purchase intention as the dependent variable, product purity perception and ingredient variety inference as mediators, based on 5000 bootstrap samples (PROCESS Model 6; Hayes, 2012). Results showed that ingredient variety inference did not mediate the effect of packaging colorfulness on purchase intention (b = −.01, SE = .06; 95% CI: [-.1226, .1138]), but product purity perception mediated (b = .40, SE = .10; 95% CI: [.2137, .6009]). As Fig. 4 indicated, ingredient variety inference has a negative impact on product purity perception (b = −.27, SE = .05, t = −5.85, p < .001), and product purity perception positively affects purchase intention (b = .62, SE = .07, t = 8.89, p < .001). The confidence interval of the indirect effects through ingredient variety inference to product purity perception and subsequently to purchase intention excluded zero, which confirmed a significant serial mediation (b = .18, SE = .05; 95% CI: [.0944, .2939]). The direct effect of packaging colorfulness on purchase intention was not significant (b = .07, SE = .14, t = .49, p = .625). These results indicated that the relationship between packaging colorfulness and purchase intention is fully mediated by product purity perception and ingredient variety inference.

8.4. Discussion

Study 4 utilized single- (vs, multi-) colored packaging to test behavioral consequences of the colorfulness-purity effect. We showed that consumers were more willing to purchase a product when it was packaged with less colorful patterns. This is because less colorful packaging triggered lower ingredient variety inference and thus enhanced higher perceived product purity.

9. General discussion

Considering the significant impact of product packaging color on consumer perceptions, our study, comprising five experiments, investigates the relationship between packaging colorfulness and perceived product purity. Specifically, Studies 1A and 1B examined the basic effect, showing that consumers perceive packaging with low (vs. high) colorfulness as containing purer products, thus supporting H1.



*p < .05, **p < .01, ***p < .001.

Fig. 4. Serial mediation on purchase intention (Study 4).

Study 2 investigated the mechanism underlying this effect, revealing that inferences about ingredient variety drive the observed effect, thus supporting H2. Furthermore, we identified a boundary condition for the colorfulness effect. Specifically, Study 3 demonstrated that when ingredient information is salient, the effect of packaging colorfulness on purity perception is mitigated, thus supporting H3.

9.1. Theoretical contributions

First, our findings contribute to the literature on packaging colors. Prior research has extensively examined how color hue (Grandi and Cardinali, 2022; Motoki et al., 2023; Plasek et al., 2020), saturation (Buechel and Townsend, 2018; Hagtvedt and Adam Brasel, 2017; Liu et al., 2022; Mead and Richerson, 2018; Wang et al., 2020), and lightness (Hagtvedt, 2020; Motoki et al., 2019; Sunaga et al., 2016; Tijssen et al., 2017; Walker et al., 2010) shape consumers' judgments of product attributes. However, most of these studies have primarily concentrated on the effects of individual color attributes on consumers, with limited exploration of color effects from a broader, holistic perspective. This study focuses on colorfulness in product packaging, which is a variable about overall color perception, thereby expanding the existing literature on color.

Second, our research contributes to the literature on product purity perception. Product purity is a key marketing focus, especially in food advertising, where brands frequently emphasize the high purity of their products (Amos et al., 2021; O'Hagan, 2020). Surprisingly, marketing research has largely overlooked the drivers of product purity perception. To the best of our knowledge, existing studies have only examined the effects of brand name fluency (Cho, 2019) and packaging complexity (Ton et al., 2024) on purity perception, while our study identifies a novel visual cue that affects consumers' perceived purity. Specifically, our findings indicate that lower colorfulness in product packaging enhances consumers' purity perception.

Third, the theory of the colorfulness effect as a spillover from a sense of variety advances our understanding of spillover effects. Despite there being substantial research on spillover effects, most studies concentrated on areas such as brand extensions (Balachander and Ghose, 2003; Lei et al., 2008; Pontes and Pontes, 2021), advertising placement (Krijestorac et al., 2020; Sahni, 2016), endorsement (Carrillat et al., 2014; Johnson et al., 2022), and word of mouth (Chae et al., 2017; Sanchez et al., 2020). Limited research has examined the spillover effects of visual attributes on product perception (Bettels and Wiedmann, 2019; Hagtvedt and Patrick, 2008; Huettl and Gierl, 2012). Building on spillover effects, our research suggests that the sense of variety induced by packaging colorfulness can extend to consumers' judgments of the variety of ingredients inside, thereby contributing to the spillover effect literature.

Finally, we investigate the mediating role of inferred ingredient variety, thus expanding the literature on variety perception. Previous research has examined how horizontal and vertical displays (Deng et al., 2016), category presence (Mogilner et al., 2008), and visual and verbal depictions (Townsend and Kahn, 2014) influence consumers' variety perception. This study provides evidence that packaging featuring a low colorfulness pattern triggers higher inferences of ingredient variety, which enhances consumers' perceived product purity. These findings enrich our understanding of variety perception research from the perspective of packaging color. Notably, our study also offers the potential to integrate existing findings. For example, Song et al. (2022) explored the relationship between colorfulness and variety perception; however, our study differs in key ways. In addition to focusing on different aspects—logo colorfulness in their study versus packaging colorfulness in ours—their research, like other variety perception studies, examines product variety at the macro level, whereas we focus on the micro level of ingredient variety within products. Furthermore, our study extends their findings by exploring how variety inference influences purity perception, thereby addressing the gap in understanding how colorfulness affects consumers' perceived product purity.

9.2. Managerial Implication

Packaging plays a crucial role in marketing communication, capturing consumers' attention and shaping their understanding of product value and overall consumption experience (Krishna et al., 2017). Our study offers new insights for marketers on how to leverage packaging colorfulness to shape consumers' purity perceptions. Packaging colorfulness, as a key design element, offers distinct advantages in marketing communication. On one hand, integrating colorfulness into packaging design is a natural and unobtrusive practice, as it is widely used across various products, allowing consumers to easily encounter packaging with varying degrees of colorfulness. On the other hand, colorfulness serves as a convenient and cost-saving tool, enabling marketers to influence consumers' perceived product purity simply by altering packaging colorfulness, without changing the actual purity.

Additionally, we highlight the boundary condition of packaging colorfulness' impact on perceived purity, indicating that marketers must consider when this strategy is effective in influencing consumer judgments. Study 4 indicated that when ingredient information is salient, consumers are less likely to use packaging colorfulness to infer ingredient variety. This occurs because consumers tend to rely more on objective product information than on packaging colors when making judgments. Importantly, ingredient information can be presented in both factual forms (e.g., detailed ingredient lists or GDA/traffic light systems) (Ikonen et al., 2020) and implied forms (e.g., nutrition claims such as "rich in vitamin C″ or "low in fat") (Belei et al., 2012; Hallez et al., 2023). Therefore, marketers must be attuned to the alignment between packaging colorfulness and ingredient information, recognizing that colorfulness only influences purity perception when ingredient details are not salient.

9.3. Limitations and future research

First, real-world packaging design involves various visual elements beyond color; however, our study focused solely on colorfulness. Future research could explore how combining colorfulness with other packaging elements—such as materials, shapes, weights, brand logos, and other design features—affects consumers' perceived purity.

Second, one limitation of this study is that it focused solely on the effect of a single brand's packaging colorfulness on perceived purity. Future research could extend this study by exploring whether the colorfulness of packaging from other brands also affects the perceived product purity of the focal brand. Since supermarkets often display products from various brands side by side on shelves, with consumers easily encountering a variety of colorful packaging, the spillover effect of colorfulness is likely to occur. This would shed light on whether the interaction of visual cues from competing products influences consumer evaluations in a multi-brand environment.

Third, this study focuses exclusively on the immediate impact of packaging colorfulness on consumer judgments, overlooking potential long-term effects. Therefore, a promising direction for future research is to explore consumers' attitudes toward packaging with varying levels of colorfulness during the post-purchase phase. For instance, whether consumers perceive a discrepancy between actual and perceived product purity post-consumption, and how they react to such discrepancies. Additionally, it would be valuable to examine whether the effect of packaging colorfulness on purity perception persists after consumption experiences involving biased judgments.

Finally, our study did not account for the external environment in which product packaging is placed, such as the advertising context and store environment. Therefore, future research should further investigate whether the colorfulness effect is contingent on external environmental factors, such as advertising context and store decor. For instance, future studies could explore whether the perceived product purity is influenced

by packaging colorfulness when the packaging is placed against advertising backgrounds or store walls with varying levels of colorfulness.

CRediT authorship contribution statement

Zhiyuan Huang: Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Xiaohe Dai:** Writing – review & editing, Writing – original draft, Resources, Project administration, Data curation, Conceptualization.

Declaration of competing interest

None.

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Appendix. Study Stimuli

Study 1A: Black Chocolate.

Low Colorfulness Condition



High Colorfulness Condition



Study 1B: Soy Milk.

Low Colorfulness Condition



High Colorfulness Condition



Study 2: Shampoo.

Low Colorfulness Condition

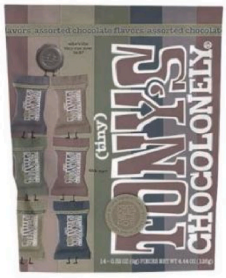


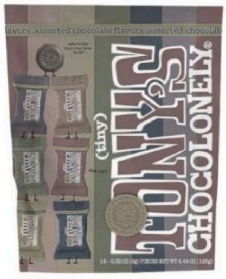
High Colorfulness Condition



Study 3: Chocolate.

Low Colorfulness





Ingredients: Sugar, cocoa mass, dry whole milk, cocoa butter, almond 1.4%, nougat 1.4% (sugar, glucose syrup, almonds 0.3%, honey 0.03%, cocoa butter, potato starch, egg white, natural vanilla extract), hazelnut 1.4%, caramel 1.4% (sugar, glucose syrup, butter, cream), pretzel pieces 1% (wheat flour, salt, sunflower oil, malted wheat flour, yeast, acidity regulator: sodium hydroxide), caramel 0.4% (sugar, glucose syrup), 0.15% sea salt, emulsifier: soya lecithin.

High Colorfulness





Ingredients: Sugar, cocoa mass, dry whole milk, cocoa butter, almond 1.4%, nougat 1.4% (sugar, glucose syrup, almonds 0.3%, honey 0.03%, cocoa butter, potato starch, egg white, natural vanilla extract), hazelnut 1.4%, caramel 1.4% (sugar, glucose syrup, butter, cream), pretzel pieces 1% (wheat flour, salt, sunflower oil, malted wheat flour, yeast, acidity regulator: sodium hydroxide), caramel 0.4% (sugar, glucose syrup), 0.15% sea salt, emulsifier: soya lecithin.

Study 4: Milk.

Low Colorfulness Condition *High Colorfulness Condition*



Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jretconser.2025.104283.

Data availability

Data will be made available on request.

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